# MMBMPHcw1

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# Mathematical Modelling in Biology, Medicine & Public Health CW1

## Appendix A

### Q4 (iii)

Below I execute the method I proposed and described in my written answer to Q4(iii) for estimating the identifiable parameters.

```
data <- read.csv("data.csv")
colnames(data) <- c("t", "y_squared")

model_function <- function(t, lambda1, lambda2) {
  ((exp(lambda1 * t) - exp(lambda2 * t)) / (lambda1 - lambda2))^2
}

# initial starting guesses for the algorithm: lambda1=0.5 (growth rate, positive); lambda2=-0.1
(decay rate, negative)
fit <- nls(y_squared ~ model_function(t, lambda1, lambda2),
           data = data,
           start = list(lambda1 = 0.5, lambda2 = -0.1))

summary(fit)
```

```
##
## Formula: y_squared ~ model_function(t, lambda1, lambda2)
##
## Parameters:
##         Estimate Std. Error t value Pr(>|t|)
## lambda1  0.20973    0.01123   18.67   <2e-16 ***
## lambda2 -0.51037    0.04785  -10.67   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.184 on 398 degrees of freedom
##
## Number of iterations to convergence: 10
## Achieved convergence tolerance: 6.981e-07
```

# Q4 (iv)

First I fit an exponential model, again using NLLS.

```
# starting values: c represents a growth rate, so  I start it near the lambda1 value found in (i
ii)
exp_fit <- nls(y_squared ~ a + b * exp(c * t),
               data = data,
               start = list(a = 0, b = 0.1, c = 0.2))

summary(exp_fit)
```

```
##
## Formula: y_squared ~ a + b * exp(c * t)
##
## Parameters:
##   Estimate Std. Error t value Pr(>|t|)
## a -1.64823    1.16023  -1.421    0.156
## b  2.06871    0.49213   4.204 3.25e-05 ***
## c  0.41392    0.02963  13.968  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.179 on 397 degrees of freedom
##
## Number of iterations to convergence: 9
## Achieved convergence tolerance: 1.158e-07
```

Next I fit a polynomial of degree 2 using the linear model function.

```
# y_squared = B0 + B1*t + B2*t^2
poly_fit <- lm(y_squared ~ poly(t, 2, raw = TRUE), data = data)

summary(poly_fit)
```

```
##
## Call:
## lm(formula = y_squared ~ poly(t, 2, raw = TRUE), data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -23.670  -3.449  -0.559   3.055  33.686
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               4.06003    1.09602   3.704 0.000242 ***
## poly(t, 2, raw = TRUE)1 -3.53476    0.63111  -5.601 3.99e-08 ***
## poly(t, 2, raw = TRUE)2  1.16944    0.07621  15.346  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.27 on 397 degrees of freedom
## Multiple R-squared:  0.8026, Adjusted R-squared:  0.8016
## F-statistic:   807 on 2 and 397 DF,  p-value: < 2.2e-16
```

Plotting the fittings:

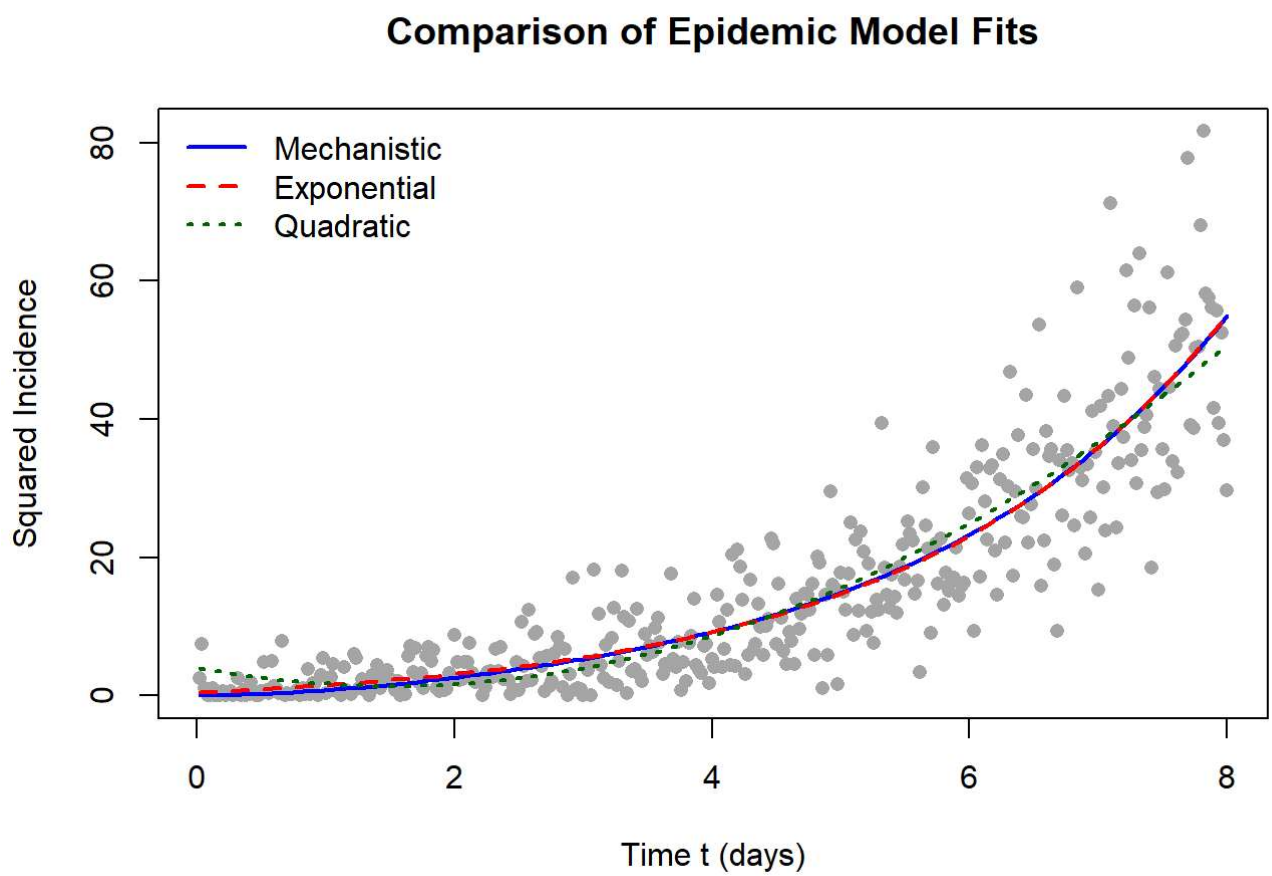
```
t_seq <- data.frame(t = seq(min(data$t), max(data$t), length.out = 100))

pred_mech <- predict(fit, newdata = t_seq)
pred_exp  <- predict(exp_fit, newdata = t_seq)
pred_poly <- predict(poly_fit, newdata = t_seq)

plot(data$t, data$y_squared,
main = "Comparison of Epidemic Model Fits",
xlab = "Time t (days)",
ylab = "Squared Incidence",
pch = 16, col = "darkgray")

lines(t_seq$t, pred_mech, col = "blue", lwd = 2, lty = 1)
lines(t_seq$t, pred_exp, col = "red", lwd = 2, lty = 2)
lines(t_seq$t, pred_poly, col = "darkgreen", lwd = 2, lty = 3)

legend("topleft",
legend = c("Mechanistic", "Exponential", "Quadratic"),
col = c("blue", "red", "darkgreen"),
lwd = 2, lty = c(1, 2, 3), bty = "n")
```


Comparison of Epidemic Model Fits
Mechanistic
Exponential
Quadratic
Squared Incidence
0
20
40
60
80
0
2
4
6
8
Time t (days)